BDCOM P3310B EPON Installation Manual



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Chapter 1 Introduction of BDCOM P3310B EPON Access Device

The document describes the characteristics and parameters of BDCOM P3310B EPON and gives an overview of BDCOM P3310B EPON.

1.1 Standard Configuration

BDCOM P3310B EPON has 4 EPON SFP ports, 2 GE optical/electric combo ports, 2 GE RJ45 electric ports, 2 GE SFP optical ports, 1 console port and 1 GE network management port. For details, see the following table.

Table 1- 1 Attributes of the necessary port

Port	Attribute
Stands for the EPON port.	EPON port, SFP port: having LINK/ACT indicators
1000M optical/electrical port	Electric port: 1000M rate, MDI/MDIX auto-identification, UTP(RJ45) port and the LINK/ACT indicators SFP port: having LINK/ACT indicators
ETH	Electric port: having the 100M rate and the LINK/ACT indicators
Console port	An RJ45 port with a rate of 9600 bps

Moreover, BDCOM P3310B EPON has two power-source sockets at its back terminal.

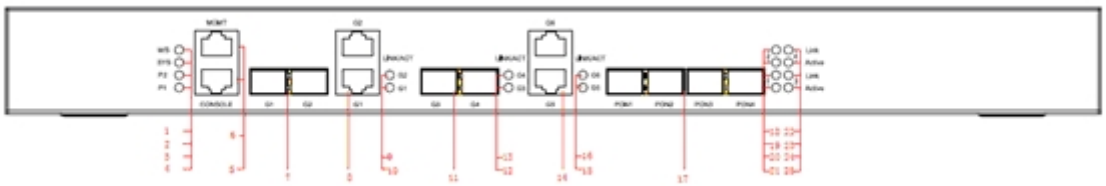


Figure 1-1 Front faceplate of BDCOM P3310B EPON

Table 1- 2 Parts at the front faceplate of BDCOM P3310B EPON

No.	Abbrev.	Name	Remarks
1	M/S	Indicator of ring network	The LED is on when it is the master station and off when it is the secondary station.
2	SYS	LED of system's running	If the indicator is always on, the system is being started.

			If the indicator flickers, the system works normally.
3	P2	LED of power source 2	When PWR2 supplies power, this LED is on.
4	P1	LED of power source 1	When PWR1 supplies power, this LED is on.
5	Console	Console port	Realizes the local management of the EPON access device.
6	MGMT	Network management port	Realizes the upgrade of the EPON access device.
7	G1，G2	GE combo optical port	Forwards the 1000M Ethernet optical signals.
8	G1，G2	GE combo electric port	Forwards the gigabit-Ethernet electrical signals.
9	G2 Link/ACT	LED of the G2 port	If the indicator is always on, the link on the port is normal. If the indicator flickers, the data is received or transmitted through the port.
10	G1Link/ACT	LED of the G1 port	If the indicator is always on, the link on the port is normal. If the indicator flickers, the data is received or transmitted through the port.
11	G3,G4	Gigabit SFP optical port	Forwards the 1000M Ethernet optical signals.
12	G3 Link/ACT	LED of the G3 port	If the indicator is always on, the link on the port is normal. If the indicator flickers, the data is received or transmitted through the port.
13	G4 Link/ACT	LED of the G4 port	If the indicator is always on, the link on the port is normal. If the indicator flickers, the data is received or transmitted through the port.
14	G5,G6	Gigabit RJ45 electric port	Forwards the gigabit-Ethernet electrical signals.
15	G5 Link/ACT	LED of the G5 port	If the indicator is always on, the link on the port is normal. If the indicator flickers, the data is received or transmitted through the port.

16	G6 Link/ACT	LED of the G6 port	If the indicator is always on, the link on the port is normal. If the indicator flickers, the data is received or transmitted through the port.
17	PON1,PON2, PON3,PON4	Gigabit OLT PON port	Realizes the access of EPON ONU.
18	PON2 LINK	LED of PON2 LINK	If this LED is on, it means that ONU is connected.
19	PON2 ACT	LED of PON2 ACT	If this LED flickers, it means that the data is being transmitted.
20	PON1 LINK	LED of PON1 LINK	If this LED is on, it means that ONU is connected.
21	PON1 ACT	LED of PON1 ACT	If this LED flickers, it means that the data is being transmitted.
22	PON4 LINK	LED of PON4 LINK	If this LED is on, it means that ONU is connected.
23	PON4 ACT	LED of PON4 ACT	If this LED flickers, it means that the data is being transmitted.
24	PON3 LINK	LED of PON3 LINK	If this LED is on, it means that ONU is connected.
25	PON3 ACT	LED of PON3 ACT	If this LED flickers, it means that the data is being transmitted.

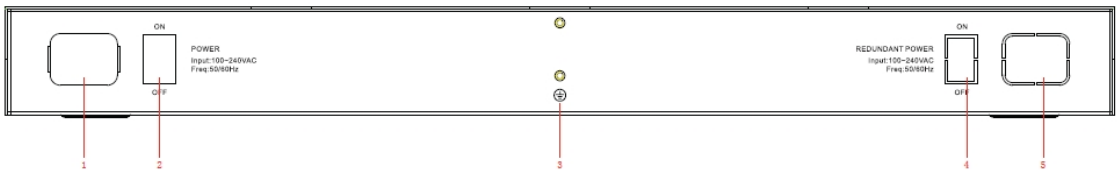


Figure 1-2 Back faceplate of BDCOM P3310B EPON

1.2 Parts at the Rear Faceplate of P3310B EPON Access Device

No.	Abbrev.	Name	Remarks
1	None	AC power socket	AC100-240V
2	None	Power switch	**ON** means opening the power, while **OFF** means cutting off the power.

3	None	Grounding column	The grounding must be fine.
4	None	Backup power on-off	Pressing “ON” means opening power, while pressing “OFF” means cutting off the power. It is used for backup power.
5	None	Standby AC power socket	AC100～240V, used as the backup power

1.3 Attribute Parameters of BDCOM P3310B EPON

Protocol standard	IEEE 802.1d Spanning Tree Protocol IEEE 802.1p Class of Service IEEE 802.1q tagged VLAN IEEE 802.3x Flow control IEEE 802.3ad Link aggregation IEEE 802.3ah Ethernet in the First Mile Task Force
Standard of IP routing protocol	RFC 1058 RIP RFC 1723 RIP v2 RFC 1583 OSPF v2
Network management standard	RFC 1157 SNMP v1/v2 RFC 1213 MIB II RFC 1757 RMON 1,2,3,9
Memory	EPROM: 512 Bytes Flash Memory: 8M Bytes SDRAM: 128MBytes
Standard configuration	One Console port 1 gigabit network management port 2 gigabit electric/optical combo ports 2 gigabit electric ports 2 gigabit optical ports 4 gigabit PON ports
Specifications	442*315*44MM
Working temperature/humidity	0℃-60℃; 10%-85% no condensation
Storage temperature/ humidity	-40℃-80℃; 5%-95% no condensation
AC/DC power supply	Input voltage: AC100-240V Input frequency: 47-63Hz Input current: 1A/230V
Power consumption	Up to 48W

1.4 ROHS Description

Parts	Toxic or harmful substances or elements					
	Pb	Hg	Cd	Cr(VI)	PBB	PBDE
Chasis	O	O	O	O	O	O
Rack	O	O	O	O	O	O
Baseboard	O	O	O	O	O	O
Module	O	O	O	O	O	O
Interface Card	O	O	O	O	O	O

O: The toxic or harmful substances' levels in each homogeneous materials of each part, are under the limitation of SJ/T 11363—2006 regulation

X: The toxic or harmful substances' levels at least in one homogeneous materials of one part, exceed the limitation of SJ/T 11363—2006 regulation



Chapter 2 **Installation Preparation**

2.1 Cautions

Similar to other electronic products, the semiconductor chip easily gets damaged if you power on and off abruptly and frequently. To restart up the switch of BDCOM P3310B EPON, you have to open the power on-off three or five seconds after the power is cut off.

Avoid severe collision or falling down from the height to protect the parts in BDCOM P3310B EPON.

Use correct outside ports to connect BDCOM P3310B EPON. Do not insert the Ethernet plug into the console port (RJ45 8-line socket). Similarly, do not insert the console cable into the console port (RJ45 8-line socket).

Note:

1) When you plug or dial out the power line, keep the power line horizontal with the power socket.

2) When the lifetime of our products ends, handle them according to national laws and regulations, or send these products to our company for collective processing.

2.2 Safety Advice

2.2.1 Safety Principles

- Keep dustless and clean during or after the installation.
- Put the cover at the safe place.
- Put tools at the right place where they are not easily falling down.
- Put on relatively tight clothes, fasten the tie or scarf well and roll up the sleeve, avoiding stumbling the chassis.
- Put on the protective glasses if the environment may cause damage to your eyes.
- Avoid incorrect operations that may cause damage to human or devices.

2.2.2 Safety Notices

The safety notices mentioned here means that improper operation may lead to body damage.

- Read the installation guide carefully before you operate the system.

- Only professionals are allowed to install or replace the EPON access device.

- Pull out the AC power socket and close the direct-current power before operating on the chassis or working beside the power source.

- The final configuration of products must comply with relative national laws and regulations.

2.2.3 Safety Principles for Live Working

When you work under electricity, following the following principles:

- Put off ornaments, such as ring, necklace, watch and bracelet, before you operate under live working. When metal articles connect the power to the ground, short circuit happens and components may be damaged.

- Pull out the AC power socket and close the direct-current power before operating on the chassis or working beside the power source.

- When the power is on, do not touch the power.

- Correctly connect the device and the power socket.

- Only professionals are allowed to operate and maintain the device.

- Read the installation guide carefully before the system is powered on.

Note:

1) Check potential dangers, such as the humid floor, ungrounded extensible power line and tatty power line.

2) Install the emergent on-off at the working room for turning off the power when trouble happens.

3) Turn off the power on-off of the EPON access device and plug off the power line before installing or uninstalling the machine box or working beside the power.

4) Do not work alone if potential dangers exist.

5) Cut off the power before checkout.

6) If trouble happens, take the following measures:

   A. Cut off the system's power.

   B. Alarm.

   C. Take proper measures to help persons who are hit by the disaster. Artificial respiration is needed if necessary.

   D. Seek for medical help, or judge the loss and seek for available help.

2.2.4 Electrostatic Discharge Prevention

Electrostatic discharge may damage devices and circuits. Improper treatment may cause this device to malfunction completely or discontinuously.

Move or locate the devices according to the measures of electrostatic discharge prevention, ensuring the chassis connects the ground. Another measure is to wear the static-proof hand ring. If there is no hand ring, use the metal clip with the metal cable to clip the unpainted metal part of the chassis. In this case, the static is discharged to the ground through the metal cable of the clip. You can also discharge the static to the ground through your body.

2.3 Requirements for Common Locations

This part describes the requirements for the installation locations.

2.3.1 Environment

The location of the chassis, cabinet planning and indoor cabling are very important for normal system's function. Short distance between devices, bad ventilation and untouchable control plate will cause maintenance problems, systematic faulty and breakdown.

For location planning and device locating, refer to section 2.3.2 "Location Configuration Prevention".

2.3.2 Location Configuration Prevention

The following preventive measures assist you to design the proper environment for this EPON access device.

- Make sure that the workshop is well-ventilated, the heat of electrical devices is well-discharged and sufficient air circulation is provided for device cooling.
- Avoid to damage devices by following the electrostatic discharge prevention procedure.
- Put the chassis at the place where cool air can blow off the heat inside the chassis. Make sure the chassis is sealed because the opened chassis will reverse the cool air flow.

2.3.3 Cabinet Configuration

The following content assists you to make a proper cabinet configuration:

- Each device on the cabinet gives off heat when it runs. Therefore, the sealed cabinet must have the heat-discharge outlet and the cooling fan. Do not put the devices too close, avoiding bad ventilation.
- When you install the machine box at the open cabinet, prevent the frame of the cabinet from blocking the airway of the machine box.

- Ensure that nice ventilation is provided for the devices installed at the bottom of the cabinet.

- The clapboard separates exhaust gas and inflow air, and boost the cool air to flow in the chassis. The best location of the clapboard is decided by the air flow mode in the chassis, which can be obtained through different location tests.

2.3.4 Power Requirements

Make sure that the power supply has nice grounding and the power at the input side of this EPON access device is reliable. The voltage control can be installed if necessary. At least a 240 V, 10A fuse or a breaker is provided in the phase line if you prepare the short-circuit prevention measures for a building.

Caution:

If the power supply system does not have good grounding, or the input power disturbs too much and excessive pulses exist, the error code rate of communication devices increases and even the hardware system will be damaged.

2.4 Installation Tools and Device

The tools and devices to install BDCOM P3310B EPON are not provided by the BDCOM P3310B EPON. You yourself need to prepare them. The following are the tools and devices needed for the typical installation of BDCOM P3310B EPON:

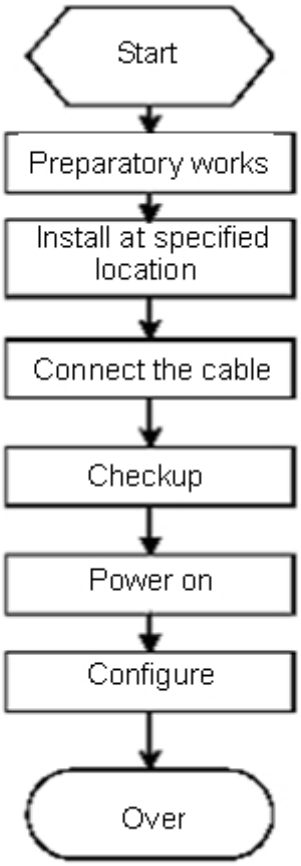
- Screwdriver
- Static armguard
- Bolt
- Ethernet cable
- Other Ethernet terminal devices
- Control terminal

Chapter 3 Installing P3310B EPON Device

Caution:

Only professionals are allowed to install or replace the devices of this device.

3.1 Installation Flow of BDCOM P3310B EPON



3.2 Installing the Machine Box of BDCOM P3310B EPON

The machine box of BDCOM P3310B EPON can be installed on the desk or can be fixed to other cabinets. Your network installation requirements can be met if you conduct the operations according to the following procedure. It can be described in the following two parts:

- Installing the Machine Box on the Desk
- Installing the Chassis on the Cabinet

3.2.1 Installing the Machine Box on the Desk

BDCOM P3310B EPON can be directly put on the smooth and safe desk.

Note:

Do not put items with more than 4.5kg on BDCOM P3310B EPON, or it will be damaged.

3.2.2 Installing the Chassis on the Cabinet

The machine box of BDCOM P3310B EPON is fixed on the cabinet through the brackets. When you fix the brackets, the front faceplate of BDCOM P3310B EPON faces forward. The detailed operations are shown in Figure 3-1.

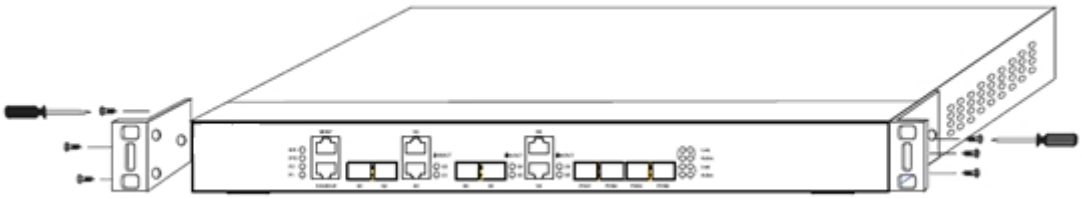


Figure 3-1 Fixing the machine box of BDCOM P3310B EPON to the cabinet

After the brackets are installed, install the router on the cabinet. See the following figure:

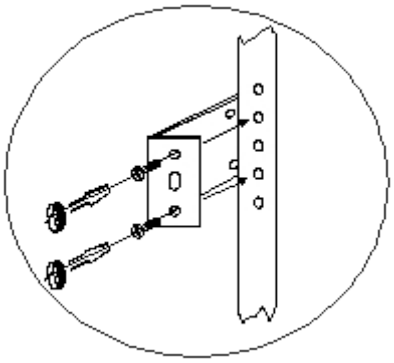


Figure 3- 2 Installing the switch on the cabinet

3.3 Connecting the Port

3.3.1 Connecting the Console Port

BDCOM P3310B EPON has a Console port.

The rate of the console port is a value ranging from 1200bps to 115200bps. It has a standard RJ45 plug. After you connect the console port to the serial port of PC through a console cable, you can configure and monitor BDCOM P3310B EPON by running a terminal emulation software, such as super Windows terminal. The cable is provided according to the host. The communication parameters of the terminal serial port can be set to a rate of 9600bps, eight data bits, one stop bit, no sum check bit and traffic control.

The RJ45 connector of the console port is shown in the following figure. The RJ45 plug corresponds to the RJ45 socket, whose pins can be aligned from left to right with the value from 1 to 8.

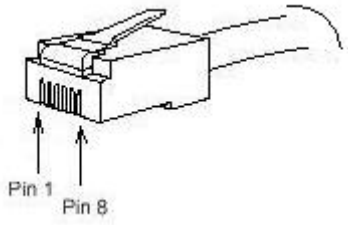


Figure 3- 3 RJ-45 connector of the console port

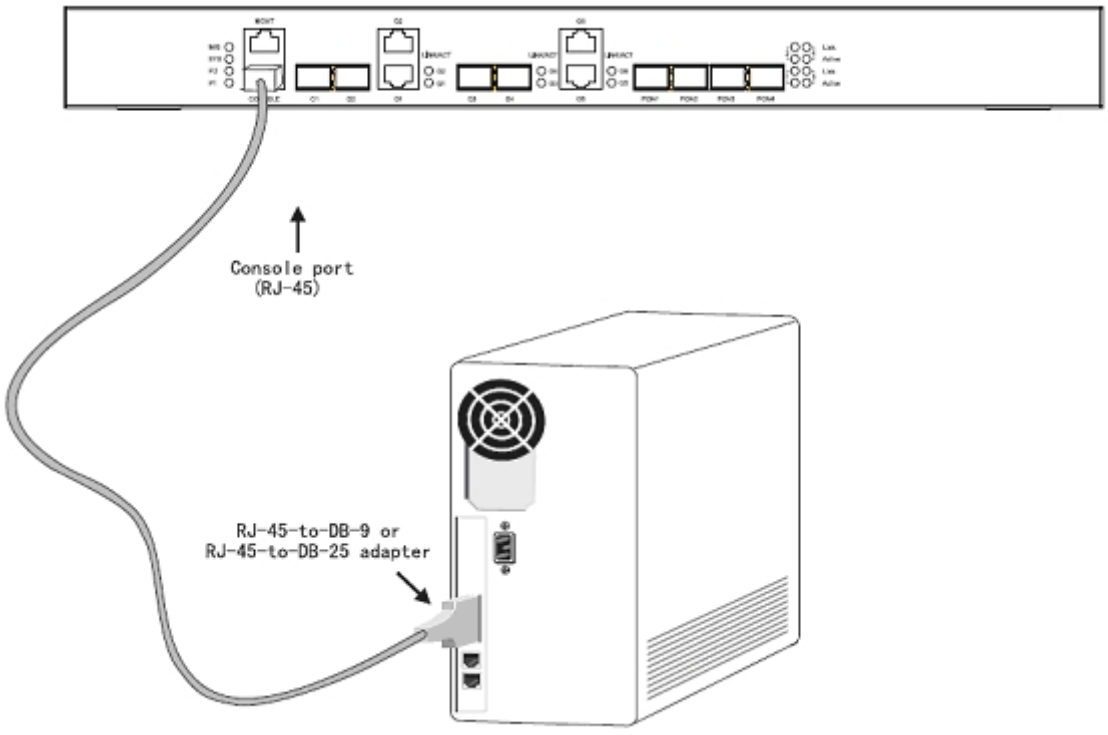


Figure 3-4 Connecting the console port of BDCOM P3310B EPON and the computer

Table 3- 1 Definition of the pins of the UTP port

No.	Name	Symbol	Remarks

1	Carrier Detecting	CD	No connect
2	Data receiving	RXD	Input
3	Data-line device ready	DSR	No connect
4	Data transmitting	TXD	Output
5	Transmission requesting	RTS	No connect
6	Response transmitting	CTS	No connect
7	Data terminal ready	DTR	No connect
8	Signal ground	SG	GND

Note:

The console port of BDCOM P3310B EPON does not support traffic control. Therefore, you must set the option **data traffic control** to **none** when you configure BDCOM P3310B EPON with the super terminal. Otherwise, the single-pass problem will arise on the super terminal.

Otherwise, the single-pass problem will arise on the super terminal. The cable is used to connect the console port of BDCOM P3310B EPON and the outside console terminal device. One end of the cable is a 8-pin RJ45 plug and the other end is a 9-hole plug (DB9). The RJ45 plug is put into the socket of the console port on BDCOM P3310B EPON. The inner line connection in the cable is shown in figure 3-1. The console cable is numbered as RLC0301.

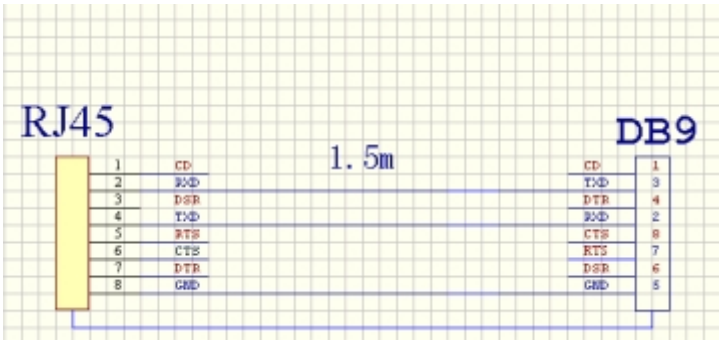


Figure 3- 5 Cable connection at the console port

3.3.2 Connecting the EPON SFP Interface

BDCOM P3310B EPON has 4 EPON SFP interfaces. The indicators lie in the right side of these interfaces, indicating the LINK/ACT state. They are used to connect the EPON access devices and other ONUs.

A single-mode SC optical fiber and an EPON optical module are needed for connect BDCOM P3310B EPON and other ONUs.

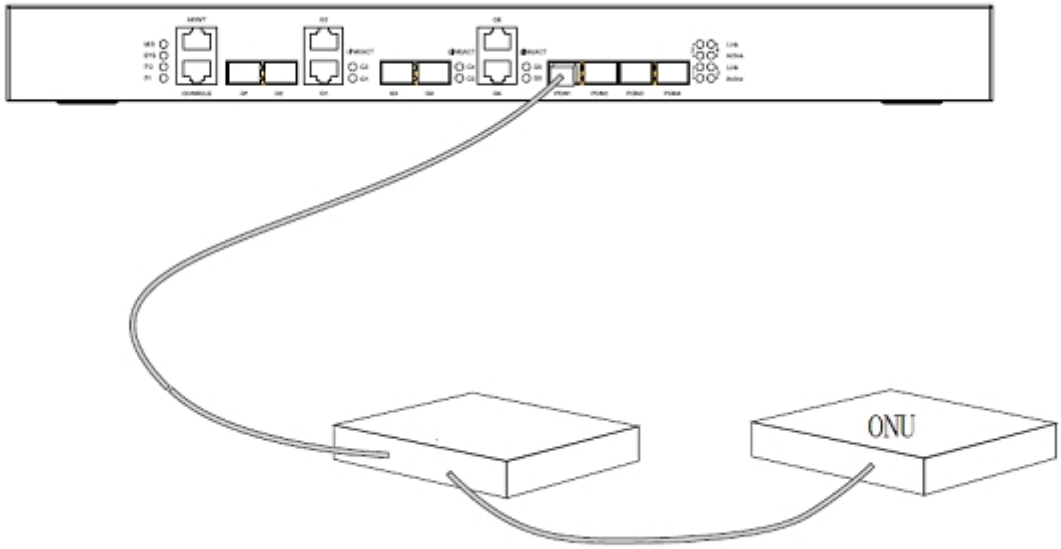


Figure 3-6 Connecting the EPON interface and other ONUs or optical splitters

3.3.3 Connecting Ethernet-1000M Optical Port

BDCOM P3310B EPON provides four 1000M optical SFP ports. Each port has a corresponding indicator for showing the LINK/ACT state of the port. If the indicator is always on, the port is normally linked; if the indicator flickers, the data is forwarded through the port. To use the optical port, you need connect it to the SFP optical module, and then to other Ethernet terminal devices through an optical fiber.

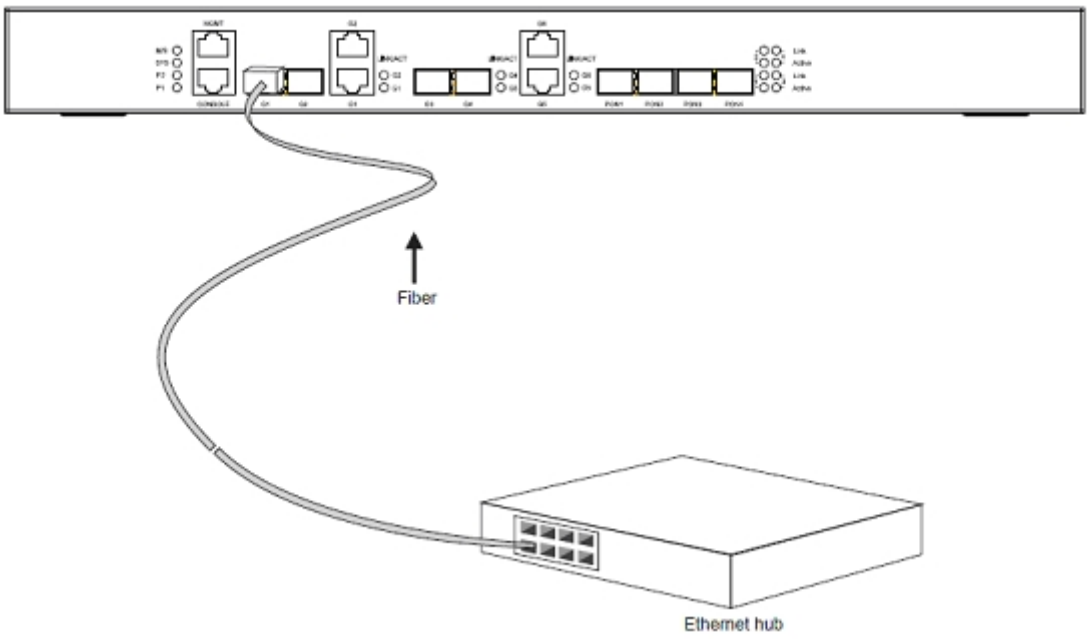


Figure 3-7 Connecting the 1000M optical port and other switches

3.3.4 Connecting the 1000M Ethernet Port

BDCOM P3310B EPON provides four fixed 1000M optical/electrical Ethernet port. The indicators on the right side are LINK/ACT lights.

To use the optical port, you need insert the SFP module to the port and then connect other Ethernet devices through the optical fiber cable. If the port is inserted with the SFP module, the corresponding electrical port cannot be used.

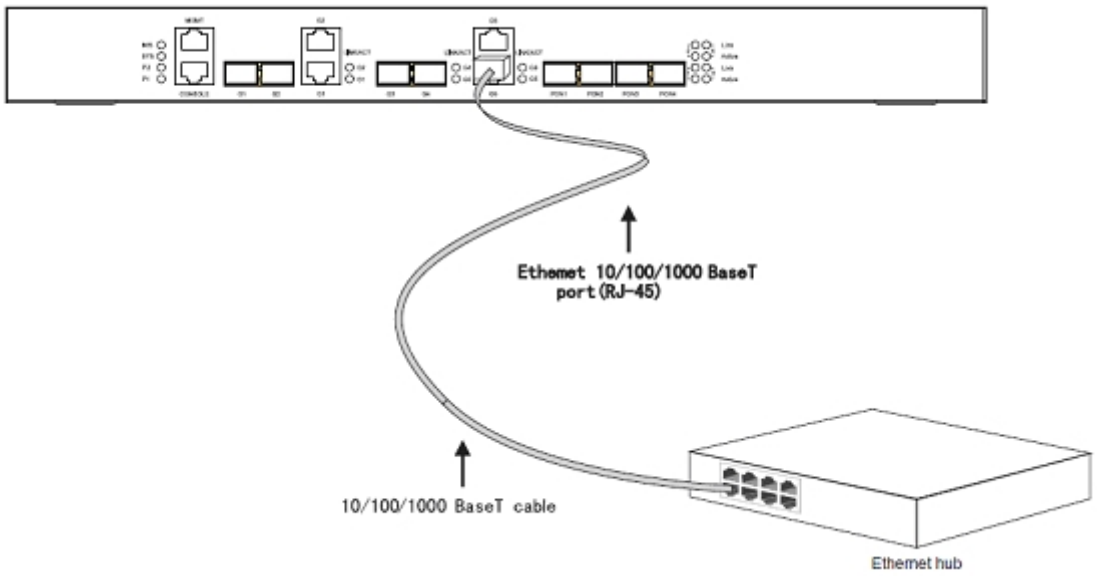


Figure 3-8 Connecting the 1000M optical port and other devices

To use the electrical port, you can connect the electric port and other Ethernet devices with the direct-through or cross cable through the UTP port of the EPON access device. In this case, the corresponding optical port cannot be used. The numbering order of the pins in the UTP port is the same as the console port.

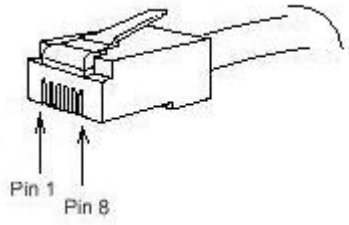


Figure 3- 9 RJ-45 connector of the console port

Because 4 1000Base-T ports of BDCOM P3310B EPON support the MDI/MDIX auto-identification of the cable, BDCOM P3310B EPON can adopt five classes of direct-through/cross network cables when it connects other Ethernet terminals.

Table 3- 3 Definition of the pins of the 1000M RJ45 port

No.	Pin Name	Symbol	Remarks

1	Sending the normal phase of the data	TXD1+	Output
2	Sending the paraphase of the data	TXD1-	Output
3	Receiving the normal phase of the data	RXD1+	Input
4	Sending the normal phase of the data	TXD2+	Output
5	Sending the paraphase of the data	TXD2-	Output
6	Receiving the paraphase of the data	RXD1-	Input
7	Receiving the normal phase of the data	RXD2+	Input
8	Receiving the paraphase of the data	RXD1-	Input

The direct-through or cross network cable has the function of auto-identification, so the five classes of direct-through/cross network cables can be used to connect other Ethernet devices.

3.4 Checking After Installation

After BDCOM P3310B EPON is installed, please conduct the following checkups before it is powered.

- If the EPON access device is installed on the cabinet, check whether the installation point between the cabinet and the EPON access device is strong. If the EPON access device is installed on the desk, check whether there is enough space for the EPON access device to discharge its heat and whether the desk is stable.
- Check whether the connected power meets the power requirements of the EPON access device.
- Check whether the grounding line of the EPON access device is correctly connected.
- Check whether the EPON access device is correctly connected to other terminal devices.

Chapter 4 Maintaining the EPON Access Device

Caution:

1) Before opening the chassis, make sure that you have released the static you carried and then turn off the power on-off of the EPON access device. Before operating any step in Appendix B, read the section “Safety Advice”.

2) Before performing operations beside the power source or on the chassis, turn off the power on-off and plug out the power cable.

4.1 Opening the Chassis

This section describes how to open the cover of the EPON access device, required tools and operation methods.

Caution:

When the power cable still connects the power source, do not touch it.

When you open the cover the EPON access device, you may use the following tools: These tools are:

- Crossed screwdriver
- Static armguard

Perform the following steps to open the cover of the EPON access device.

(1) Turn off the power on-off of the EPON access device.

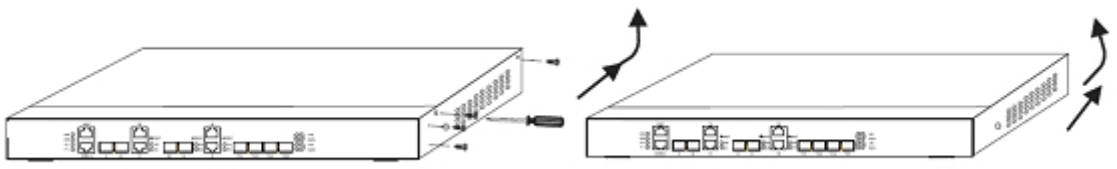
(2) Plug out all cables connected the back of EPON access device.

(3) Take out the bolt from the chassis with the screwdriver.

Note:

The chassis comprises of two parts: cover and bottom.

(4) Open the cover by holding two sides of the cover towards the direction of the arrow key shown in the following figure:



(5) When the cover is opened, put it aside. The main board of the system appears.

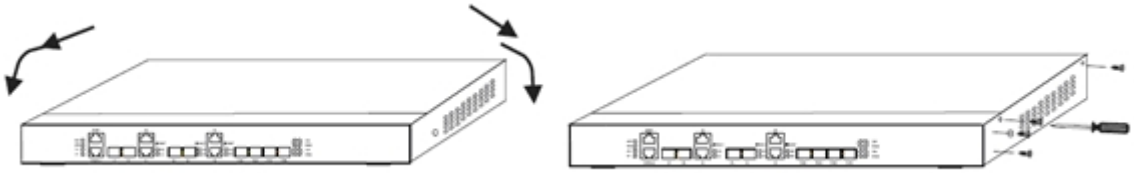
Note:

After taking off the cover, put it horizontally and avoid it to be crushed or collided. Otherwise, the chassis is hard to install.

4.2 Closing Chassis

The section mainly describes how to put the cover and close the chassis. Do as follows:

(1) Put them well according to their locations and joint them together along their sides.



(2) See the following figure.

(3) When the cover and the bottom are closely tied, let the cover slide the slot of the front template at the bottom.

(4) Nail the bolt and screw it tightly with the screwdriver.

(5) Reinstall the EPON access device on the cabinet or the desk.

(6) Reconnect all cables of the switch.

Chapter 5 **Hardware Fault Analysis**

The part describes how to remove the fault from the EPON access device.

5.1 Fault Separation

The key for resolving the systematic faults is to separate the fault from the system. You can compare what the system is doing with what the system should do to detect the fault. You need to check the following subsystems:

- Power source—providing the power
- Port, cable and connection—ports on the front template of the EPON access device and the cables connecting these ports

5.1.1 Performing the Following Power-Source Steps

- If the EPON access device is too hot, check whether the air outlet and air inlet are clean and then do relative operations in section 2.3 “Requirements for Common Locations”. The temperature of the locale where the EPON access device runs should be from minus 40 to 85 Celsius degree.
- If the EPON access device cannot be started and the PWR indicator is off, check the power.

5.1.2 Faults Relative with Port, Cable and Connection

Do the following checkups to help remove the fault:

- If the port of the EPON access device cannot be linked, check whether the cable is correctly connected and whether the peer connection is normal.
- If the console port does not work after the system is started up, check whether the console port is set to a baud rate of 9600 bps, eight data bits, no sum check bit, one stop bit and no traffic control.

5.2 Indicator Description

The LEDs are used to show what operations are conducted by the EPON access device. For the LEDs of BDCOM P3310B EPON and their description, see the following table:

No.	Abbrev.	Name	Remarks
1	PWR	Power indicator	This LED is on if the EPON access device is powered.

2	SYS	System indicator	If the indicator is always on, the system is being started. If the indicator flickers, the system works normally.
3	None	Indicator of each port	If the indicator is always on, the link on the port is normal. If the indicator flickers, the data is received or transmitted through the port.